Lab 2- Learning-SQL

In the last lab, we learned how to create, delete databases and tables, also how to read and insert data in tables.

In the dataset soap_store:

```
USE soap_store;

SHOW tables;

CREATE TABLE cats
(
cat_id INT NOT NULL PRIMARY KEY AUTO_INCREMENT,
name VARCHAR(100),
breed VARCHAR(100),
age INT
);

DESC  cats;

INSERT INTO cats
(name, breed, age)
VALUES (“Misty”, “shorthair”, 9),
(“Rover”, “longhair”, 4),
(“skippy”, “Tabby”, 6) ,
(“Jackson”, “Sphynx”, 7),
(“Ringo”, “Sphynx”, 7),
(“Cindy”, “Maine Coon”, 7),
(“Dumbledore”, “Maine Coon”, 7),
(“Egg”, “Sphynx”, 7);
```

Today we will lean:

1. UPDATE

Updating existing sets of data is done using the `UPDATE` and `SET` keywords (https://www.datacamp.com/doc/mysql/mysql-update):

```
`UPDATE table_name SET column_name='value' WHERE [some condition]`;
```

For example:

```
UPDATE cats SET age=14 WHERE name='Misty';
UPDATE cats SET breed='Shorthair' WHERE breed='Tabby';
```

BE CAREFUL, there's no undo button, so its good practice to `SELECT` the data you want to update first, before then running an `UPDATE` on it.

```
SELECT * FROM cats WHERE name='Jackson';
-- +--------+---------+--------+------+
-- | cat_id | name    | breed  | age  |
-- +--------+---------+--------+------+
-- |      7 | Jackson | Sphynx |    7 |
-- +--------+---------+--------+------+

UPDATE cats SET name='Jack' WHERE name='Jackson';
-- Query OK, 1 row affected (0.01 sec)
-- Rows matched: 1  Changed: 1  Warnings: 0

SELECT * FROM cats WHERE name='Jackson';
-- Empty set (0.00 sec)

SELECT * FROM cats WHERE name='Jack';
-- +--------+------+--------+------+
-- | cat_id | name | breed  | age  |
-- +--------+------+--------+------+
-- |      7 | Jack | Sphynx |    7 |
-- +--------+------+--------+------+
-- 1 row in set (0.01 sec)

SELECT * FROM cats WHERE name='Ringo';
-- +--------+-------+-------+------+
-- | cat_id | name  | breed | age  |
-- +--------+-------+-------+------+
-- |      1 | Ringo | Tabby |    4 |
-- +--------+-------+-------+------+
-- 1 row in set (0.00 sec)


UPDATE cats SET breed='Shorthair' WHERE name='Ringo';
-- Query OK, 1 row affected (0.01 sec)
-- Rows matched: 1  Changed: 1  Warnings: 0

SELECT * FROM cats WHERE name='Ringo';
-- +--------+-------+-----------+------+
-- | cat_id | name  | breed     | age  |
-- +--------+-------+-----------+------+
-- |      1 | Ringo | Shorthair |    4 |
-- +--------+-------+-----------+------+
-- 1 row in set (0.00 sec)
```

```
SELECT * FROM cats WHERE breed='Maine Coon';
-- +--------+------------+------------+------+
-- | cat_id | name       | breed      | age  |
-- +--------+------------+------------+------+
-- |      2 | Cindy      | Maine Coon |   10 |
-- |      3 | Dumbledore | Maine Coon |   11 |
-- +--------+------------+------------+------+
-- 2 rows in set (0.00 sec)

UPDATE cats SET age=12 WHERE breed='Maine Coon';
-- Query OK, 2 rows affected (0.01 sec)
-- Rows matched: 2  Changed: 2  Warnings: 0

SELECT * FROM cats WHERE breed='Maine Coon';
-- +--------+------------+------------+------+
-- | cat_id | name       | breed      | age  |
-- +--------+------------+------------+------+
-- |      2 | Cindy      | Maine Coon |   12 |
-- |      3 | Dumbledore | Maine Coon |   12 |
-- +--------+------------+------------+------+
```

2. DELETE

DELETE entries in a table using the `DELETE` keyword. As with UPDATE, make sure you SELECT the data first to ensure you're targeting the right entries.
Syntax:

```
DELETE FROM table_name WHERE [some condition];
```

For example:

```
SELECT * FROM cats;
-- +--------+---------------+------------+------+
-- | cat_id | name          | breed      | age  |
-- +--------+---------------+------------+------+
-- |      1 | Ringo         | Shorthair  |    4 |
-- |      2 | Cindy         | Maine Coon |   12 |
-- |      3 | Dumbledore    | Maine Coon |   12 |
-- |      4 | Egg           | Persian    |    4 |
-- |      5 | Misty         | Tabby      |   13 |
-- |      6 | George Michael | Ragdoll   |    9 |
-- |      7 | Jack          | Sphynx     |    7 |
-- +--------+---------------+------------+------+

SELECT * FROM cats WHERE name='egg';
-- +--------+------+---------+------+
-- | cat_id | name | breed   | age  |
-- +--------+------+---------+------+
```

```
-- |      4 | Egg  | Persian |    4 |
-- +--------+------+---------+------+
```

```
DELETE FROM cats WHERE name='egg';
-- Query OK, 1 row affected (0.01 sec)
```

```
SELECT * FROM cats WHERE name='egg';
-- Empty set (0.00 sec)
```

```
SELECT * FROM cats;
-- +--------+---------------+------------+------+
-- | cat_id | name          | breed      | age  |
-- +--------+---------------+------------+------+
-- |      1 | Ringo         | Shorthair  |    4 |
-- |      2 | Cindy         | Maine Coon |   12 |
-- |      3 | Dumbledore    | Maine Coon |   12 |
-- |      5 | Misty         | Tabby      |   13 |
-- |      6 | George Michael | Ragdoll    |    9 |
-- |      7 | Jack          | Sphynx     |    7 |
-- +--------+---------------+------------+------+
```

IMPORTANT: Notice how the auto-incremented cat_id column behaves when a line is deleted. All cat_ids remain the same as before, there just becomes a gap in the numbers.
BE CAREFUL, if you don't use a `WHERE` condition, you may delete all entries in the table!

DELETE FROM cats; will delete **all** entries into `cats`.
DELETE FROM cats;
Query OK, 4 rows affected (0.00 sec)

```
SELECT * FROM cats;
Empty set (0.00 sec)
```

3. String Functions

[String functions](https://dev.mysql.com/doc/refman/8.0/en/string-functions.html) allow you to manipulate string values as they're being retrieved from the tables.

First Create a table called books

```
CREATE TABLE books
(
book_id INT (20) NOT NULL AUTO_INCREMENT,
author_fname VARCHAR(30) NOT NULL DEFAULT 'NO NAME PROVIDED',
author_lname VARCHAR(30) NOT NULL DEFAULT 'NO NAME PROVIDED',
book_title VARCHAR(200) NOT NULL DEFAULT 'NO NAME PROVIDED',
PRIMARY KEY (book_id)
);
```

```
INSERT INTO books (author_fname, author_lname, book_title)
VALUES (“Jhumpa”, “Lahiri“, “The Namesake”),
(“Neil”, “Gaiman “, “Norse Mythology”),
(“Neil”, “Gaiman “, “American Gods”),
(“Jhumpa”, “Lahiri “, “Interpreter of Maladies”),
(“Dave”, “Eggers “, “A Hologram for the King: A Novel”),
(“Dave”, “Eggers “, “The Circle”),
(“Michael”, “Chabon“, “The Amazing Adventures of Kavalier & Clay”),
(“Patti”, “Smith “, “Just Kids”),
(“Dave”, “Eggers“, “A Heartbreaking Work of Staggering Genius”);
```

A few examples (there are loads that you may never need to know about)

3.1 CONCAT

Concatenates several string values together.
Syntax:

```
CONCAT(x, y, z);
SELECT CONCAT(column_name, ' ', another_column_name) FROM table_name;
-- +---------------------------------------+
-- | CONCAT(author_fname, ' ', author_lname) |
-- +---------------------------------------+
-- | Jhumpa Lahiri                         |
-- | Neil Gaiman                           |
-- | Neil Gaiman                           |
-- | Jhumpa Lahiri                         |
-- | Dave Eggers                           |
-- | Dave Eggers                           |
-- | Michael Chabon                        |
-- | Patti Smith                           |
-- | Dave Eggers                           |
-- | Neil Gaiman                           |
-- | Raymond Carver                        |
-- | Raymond Carver                        |
-- | Don DeLillo                           |
-- | John Steinbeck                        |
-- | David Foster Wallace                  |
-- | David Foster Wallace                  |
-- +---------------------------------------+
```

Notice that the header isn’t particularly user-friendly

```
-- +---------------------------------------+
-- | CONCAT(author_fname, ' ', author_lname) |
-- +---------------------------------------+
```

The header can be customized again using the `AS` keyword:

```
SELECT CONCAT(author_fname, ' ', author_lname) AS 'full name' FROM books;
-- +----------------------+
-- | full name            |
-- +----------------------+
-- | Jhumpa Lahiri        |
-- | Neil Gaiman          |
-- | Neil Gaiman          |
-- | Jhumpa Lahiri        |
-- | Dave Eggers          |
-- | Dave Eggers          |
-- | Michael Chabon       |
-- | Patti Smith          |
-- | Dave Eggers          |
-- | Neil Gaiman          |
-- | Raymond Carver       |
-- | Raymond Carver       |
-- | Don DeLillo          |
-- | John Steinbeck       |
-- | David Foster Wallace |
-- | David Foster Wallace |
-- +----------------------+
```

CONCAT_WS (concatenate with separator)

Syntax:

```
CONCAT(separator, X, Y, Z);
-- XseparatorYseparatorZ
```

```
SELECT CONCAT_WS(' - ', book_title, author_fname, author_lname) FROM books;
-- +---------------------------------------------------------------------------+
-- | CONCAT_WS(' - ', title, author_fname, author_lname)                       |
-- +---------------------------------------------------------------------------+
-- | The Namesake - Jhumpa - Lahiri                                            |
-- | Norse Mythology - Neil - Gaiman                                           |
-- | American Gods - Neil - Gaiman                                             |
-- | Interpreter of Maladies - Jhumpa - Lahiri                                 |
-- | A Hologram for the King: A Novel - Dave - Eggers                          |
-- | The Circle - Dave - Eggers                                                |
-- | The Amazing Adventures of Kavalier & Clay - Michael - Chabon              |
-- | Just Kids - Patti - Smith                                                 |
-- | A Heartbreaking Work of Staggering Genius - Dave - Eggers                 |
-- | Coraline - Neil - Gaiman                                                  |
-- | What We Talk About When We Talk About Love: Stories - Raymond - Carver |
-- | Where Im Calling From: Selected Stories - Raymond - Carver                |
-- | White Noise - Don - DeLillo                                               |
-- | Cannery Row - John - Steinbeck                                            |
```

```
-- | Oblivion: Stories - David - Foster Wallace                          |
-- | Consider the Lobster - David - Foster Wallace                        |
-- +----------------------------------------------------------------------+
```

4.2 SUBSTRING

Returns part of a string. Which part depends on the arguments passed in:
- SELECT SUBSTRING(column_name, startIndex, endIndex) FROM table_name;
  - returns characters from `column_name` between `startIndex` and `endIndex`.

OR

- `SELECT SUBSTRING(column_name, startIndex) FROM table_name;`
  - returns characters from `column_name` between `startIndex` the end of the full string.

```
SELECT SUBSTRING(book_title, 1, 8) FROM books;
-- +----------------------+
-- | SUBSTRING(title, 1,8) |
-- +----------------------+
-- | The Name              |
-- | Norse My              |
-- | American              |
-- | Interpre              |
-- | A Hologr              |
-- | The Circ              |
-- | The Amaz              |
-- | Just Kid              |
-- | A Heartb              |
-- | Coraline              |
-- | What We               |
-- | Where Im              |
-- | White No              |
-- | Cannery               |
-- | Oblivion              |
-- | Consider              |
-- +----------------------+
```

If only `startIndex` is passed (not `endIndex`), and `startIndex` is a negative number, the characters selected are counted from the end of the string.

```
SELECT SUBSTRING(book_title, 8) FROM books;
-- +-------------------------------------------+
-- | SUBSTRING(title, 8)                       |
-- +-------------------------------------------+
-- | esake                                     |
```

```
-- | ythology                                  |
-- | n Gods                                    |
-- | eter of Maladies                          |
-- | ram for the King: A Novel                 |
-- | cle                                       |
-- | zing Adventures of Kavalier & Clay        |
-- | ds                                        |
-- | breaking Work of Staggering Genius        |
-- | e                                         |
-- |  Talk About When We Talk About Love: Stories |
-- | Im Calling From: Selected Stories         |
-- | oise                                      |
-- |  Row                                      |
-- | n: Stories                                |
-- | r the Lobster                             |
-- +-------------------------------------------+
```

```
SELECT SUBSTRING(book_title, -8) FROM books;
-- +---------------------+
-- | SUBSTRING(title, -8) |
-- +---------------------+
-- | Namesake            |
-- | ythology            |
-- | can Gods            |
-- | Maladies            |
-- |  A Novel            |
-- | e Circle            |
-- | r & Clay            |
-- | ust Kids            |
-- | g Genius            |
-- | Coraline            |
-- |  Stories            |
-- |  Stories            |
-- | te Noise            |
-- | nery Row            |
-- |  Stories            |
-- |  Lobster            |
-- +---------------------+
```

4. Combining string functions:

Its possible to combine functions. When functions are combined, the inner-most function is evaluated first, and passed to its parent back up to the outer-most function.

```
SELECT CONCAT (SUBSTRING('Hello World', 1, 7 ), '...') AS my_output;
```

```
-- +------------+
-- | my_output  |
-- +------------+
-- | Hello W... |
-- +------------+
```

```
SELECT CONCAT ( SUBSTRING(book_title, 1, 10), '...' ) AS subtitle FROM books;
-- +---------------+
-- | subtitle      |
-- +---------------+
-- | The Namesa... |
-- | Norse Myth... |
-- | American G... |
-- | Interprete... |
-- | A Hologram... |
-- | The Circle... |
-- | The Amazin... |
-- | Just Kids...  |
-- | A Heartbre... |
-- | Coraline...   |
-- | What We Ta... |
-- | Where Im  ... |
-- | White Nois... |
-- | Cannery Ro... |
-- | Oblivion: ... |
-- | Consider t... |
-- +---------------+
```

5.1 REPLACE

`REPLACE(string | columnName, search, replace)`
Replaces all incidences of `search` in `string` or `column_name` with `replace`.
NOTE: REPLACE is case sensitive

Example:

```
SELECT REPLACE('Hello world', 'world', 'universe');
-- +-----------------------------------------+
-- | REPLACE('Hello world', 'world', 'universe') |
-- +-----------------------------------------+
-- | Hello universe                          |
-- +-----------------------------------------+
```

```
SELECT REPLACE(book_title, 'The', 'Das') FROM books;
-- +-------------------------------------------------+
-- | REPLACE(title, 'The', 'Das')                    |
-- +-------------------------------------------------+
```

```
-- | Das Namesake                                        |
-- | Norse Mythology                                     |
-- | American Gods                                       |
-- | Interpreter of Maladies                             |
-- | A Hologram for the King: A Novel                    |
-- | Das Circle                                          |
-- | Das Amazing Adventures of Kavalier & Clay           |
-- | Just Kids                                           |
-- | A Heartbreaking Work of Staggering Genius           |
-- | Coraline                                            |
-- | What We Talk About When We Talk About Love: Stories |
-- | Where Im Calling From: Selected Stories             |
-- | White Noise                                         |
-- | Cannery Row                                         |
-- | Oblivion: Stories                                   |
-- | Consider the Lobster                                |
-- +-----------------------------------------------------+

SELECT
  SUBSTRING (
    REPLACE(book_title, 'e', '3'),
    1,
    5
  )
AS 'short titles'
FROM books;
-- +--------------+
-- | short titles |
-- +--------------+
-- | Th3 N        |
-- | Nors3        |
-- | Am3ri        |
-- | Int3r        |
-- | A Hol        |
-- | Th3 C        |
-- | Th3 A        |
-- | Just         |
-- | A H3a        |
-- | Coral        |
-- | What         |
-- | Wh3r3        |
-- | Whit3        |
-- | Cann3        |
-- | Obliv        |
-- | Consi        |
-- +--------------+
```

5.2 REVERSE

Reverses the `selected` string.
SELECT REVERSE(book_title) FROM books;

```
-- +---------------------------------------------------+
-- | REVERSE(title)                                    |
-- +---------------------------------------------------+
-- | ekasemaN ehT                                      |
-- | ygolohtyM esroN                                   |
-- | sdoG naciremA                                     |
-- | seidalaM fo reterpretnI                           |
-- | levoN A :gniK eht rof margoloH A                  |
-- | elcriC ehT                                        |
-- | yalC & reilavaK fo serutnevdA gnizamA ehT         |
-- | sdiK tsuJ                                         |
-- | suineG gnireggatS fo kroW gnikaerbtraeH A         |
-- | enilaroC                                          |
-- | seirotS :evoL tuobA klaT eW nehW tuobA klaT eW tahW |
-- | seirotS detceleS :morF gnillaC mI erehW           |
-- | esioN etihW                                       |
-- | woR yrennaC                                       |
-- | seirotS :noivilbO                                 |
-- | retsboL eht redisnoC                              |
-- +---------------------------------------------------+
```

5.3 CHAR_LENGTH

Counts characters in a string.
SELECT author_lname, CHAR_LENGTH(author_lname) AS 'Length' FROM books;

```
-- +---------------+--------+
-- | author_lname  | Length |
-- +---------------+--------+
-- | Lahiri        |      6 |
-- | Gaiman        |      6 |
-- | Gaiman        |      6 |
-- | Lahiri        |      6 |
-- | Eggers        |      6 |
-- | Eggers        |      6 |
-- | Chabon        |      6 |
-- | Smith         |      5 |
-- | Eggers        |      6 |
-- | Gaiman        |      6 |
-- | Carver        |      6 |
-- | Carver        |      6 |
```

```
-- | DeLillo        |      7 |
-- | Steinbeck      |      9 |
-- | Foster Wallace |     14 |
-- | Foster Wallace |     14 |
-- +----------------+--------+
```

```
SELECT  CONCAT (author_lname, ' is  ', CHAR_LENGTH(author_lname), ' characters long')
AS 'length description' FROM   books;
```

```
-- +------------------------------------+
-- | length description                 |
-- +------------------------------------+
-- | Lahiri is  6 characters long       |
-- | Gaiman is  6 characters long       |
-- | Gaiman is  6 characters long       |
-- | Lahiri is  6 characters long       |
-- | Eggers is  6 characters long       |
-- | Eggers is  6 characters long       |
-- | Chabon is  6 characters long       |
-- | Smith is  5 characters long        |
-- | Eggers is  6 characters long       |
-- | Gaiman is  6 characters long       |
-- | Carver is  6 characters long       |
-- | Carver is  6 characters long       |
-- | DeLillo is  7 characters long      |
-- | Steinbeck is  9 characters long    |
-- | Foster Wallace is  14 characters long |
-- | Foster Wallace is  14 characters long |
-- +------------------------------------+
```

5.4 UPPER and LOWER

Converts strings to upper and lower case.

```
SELECT UPPER('hello world');
-- +---------------------+
-- | UPPER('hello world') |
-- +---------------------+
-- | HELLO WORLD         |
-- +---------------------+
```

```
SELECT UPPER(author_fname) FROM books;
-- +--------------------+
-- | UPPER(author_fname) |
-- +--------------------+
-- | JHUMPA             |
-- | NEIL               |
```

```
-- | NEIL              |
-- | JHUMPA            |
-- | DAVE              |
-- | DAVE              |
-- | MICHAEL           |
-- | PATTI             |
-- | DAVE              |
-- | NEIL              |
-- | RAYMOND           |
-- | RAYMOND           |
-- | DON               |
-- | JOHN              |
-- | DAVID             |
-- | DAVID             |
-- +-------------------+
```

```
SELECT LOWER(author_lname) FROM books;
-- +--------------------+
-- | LOWER(author_lname) |
-- +--------------------+
-- | lahiri             |
-- | gaiman             |
-- | gaiman             |
-- | lahiri             |
-- | eggers             |
-- | eggers             |
-- | chabon             |
-- | smith              |
-- | eggers             |
-- | gaiman             |
-- | carver             |
-- | carver             |
-- | delillo            |
-- | steinbeck          |
-- | foster wallace     |
-- | foster wallace     |
-- +--------------------+
```

```
SELECT CONCAT(UPPER(author_lname), ', ', LOWER(author_fname)) FROM books;
-- +----------------------------------------------------+
-- | CONCAT(UPPER(author_lname), ', ', LOWER(author_fname)) |
-- +----------------------------------------------------+
-- | LAHIRI, jhumpa                                     |
-- | GAIMAN, neil                                       |
-- | GAIMAN, neil                                       |
-- | LAHIRI, jhumpa                                     |
```

Task

1- Create the following table called **Employee** in a database called **Leon**

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	24000.00	0.00	0	90
101	Neena	Kochhar	NKOCHHAR	515.123.4568	1987-06-18	AD_VP	17000.00	0.00	100	90
102	Lex	De Haan	LDEHAAN	515.123.4569	1987-06-19	AD_VP	17000.00	0.00	100	90
103	Alexander	Hunold	AHUNOLD	590.423.4567	1987-06-20	IT_PROG	9000.00	0.00	102	60
104	Bruce	Ernst	BERNST	590.423.4568	1987-06-21	IT_PROG	6000.00	0.00	103	60
105	David	Austin	DAUSTIN	590.423.4569	1987-06-22	IT_PROG	4800.00	0.00	103	60
106	Valli	Pataballa	VPATABAL	590.423.4560	1987-06-23	IT_PROG	4800.00	0.00	103	60
107	Diana	Lorentz	DLORENTZ	590.423.5567	1987-06-24	IT_PROG	4200.00	0.00	103	60
108	Nancy	Greenberg	NGREENBE	515.124.4569	1987-06-25	FI_MGR	12000.00	0.00	101	100
109	Daniel	Faviet	DFAVIET	515.124.4169	1987-06-26	FI_ACCOUNT	9000.00	0.00	108	100
110	John	Chen	JCHEN	515.124.4269	1987-06-27	FI_ACCOUNT	8200.00	0.00	108	100
111	Ismael	Sciarra	ISCIARRA	515.124.4369	1987-06-28	FI_ACCOUNT	7700.00	0.00	108	100

2- Write a query to update the portion of the phone_number in the employees table, within the phone number the substring '124' will be replaced by '999'.

3- Write a query to get the details of the employees where the length of the first name greater than or equal to 8.

4- Write a query to display leading zeros before maximum and minimum salary.

5- Write a query to append '@leon.com' to email field.